

Comments to the Corresponding Author

The paper tests whether ignoring sampling uncertainty in matrix population models biases ecological inference. Using COMPADRE plant demographic data, the authors compare analyses with and without explicit propagation of uncertainty across multiple study designs. They find that sampling uncertainty has little effect on broad demographic metrics like population growth rate and on within-species climate–vital rate relationships. However, it can substantially affect certain trait-specific inferences, especially those involving mortality patterns, where boundary estimates (survival near 0 or 1) can bias results. Overall, the study concludes that ignoring sampling uncertainty is often acceptable for broad demographic questions but can lead to biased or fragile inference for specific life-history traits. While the study is methodologically robust and provides valuable insights, there are several areas where clarity, depth, and justification could be improved. Below is a detailed breakdown of areas for improvement.

Major Comments

1. Sample selection bias is insufficiently addressed

The authors note that only 26 of the 86 targeted studies (later increased to 31) could be included due to missing reporting of sample sizes. This is a notable limitation and would benefit from a more thorough assessment. In particular, studies reporting transition-specific sample sizes may differ systematically from those that do not, potentially in study design quality, species characteristics, or population structure, which could affect the generality of the results across COMPADRE. The statement that there is “no strong reason to expect systematic differences” is plausible but currently unsupported. I would encourage the authors to strengthen this point with an empirical comparison of key characteristics between included and excluded studies (e.g., matrix dimension, life form, mean survival rates, proportion of boundary estimates). This would help assess whether the analytical subset is representative of the broader dataset.

2. The boundary estimate problem is central but underexplored mechanistically

The paper’s key empirical result that point estimates of life expectancy and shape are systematically displaced from their sampling distributions due to boundary survival estimates is clearly important and well-illustrated. However, it is currently not formally analysed in a way that quantifies how this displacement varies across the dataset. The authors report that 46% of population-level matrices contain at least one stage with 100% survival, which appears to be a key driver of the bias, but the relationship between boundary prevalence and the magnitude of the observed discrepancy is not explicitly characterised. A more formal analysis would strengthen this central conclusion. For example, a regression or even a stratified or binned comparison showing how the difference between point estimates and sampling distribution estimates scales with the number of boundary stages or with the proportion of individuals in boundary survival stages would help clarify the mechanism underlying the pattern and increase the practical utility of the result for readers.

3. The variance-component framing is potentially misleading

The sampling-to-other variance ratio reported in Table 1 provides a useful summary, but its interpretation is complicated by the authors’ acknowledgement that pooling sample sizes across years may underestimate true sampling uncertainty by ignoring interannual variation in transition rates. As a result, the reported ratios for life expectancy and shape, which are already greater than 1, are likely conservative. This limitation would benefit from more prominent discussion rather than a brief note in the methods. More importantly, it raises a broader issue that is not fully explored in the manuscript: the extent to which variation currently attributed to “biological variation” in comparative analyses of ageing may instead reflect a combination of sampling

noise and temporal process variation, neither of which necessarily corresponds to stable among-population differences in mortality strategy. This point has direct implications for the senescence literature engaged with in the paper and would merit more explicit consideration in the discussion.

4. Analysis 2 conflates structural model change with uncertainty propagation

The authors note, somewhat in passing, that the point-estimate and uncertainty-aware models for recruitment in Analysis 2 differ not only in how uncertainty is incorporated but also in their likelihood structure (Gaussian on log per-capita recruitment versus Poisson on raw counts with an offset). They conclude that the two approaches are “virtually identical in practice,” but this makes it difficult to disentangle whether any similarity or differences arise from uncertainty propagation or from the change in model structure. As a result, it is challenging to attribute effects specifically to uncertainty propagation. At minimum, including a version of the point-estimate model using a Poisson likelihood (without uncertainty propagation) as an additional comparison would help isolate the effect of uncertainty from that of the structural model differences.

5. Generalisability claims exceed the scope of the data

The paper presents a practical checklist in the Discussion that is framed as broadly applicable to COMPADRE users. However, the empirical basis for these recommendations is relatively limited (31 studies), largely consisting of herbaceous perennial plant species. While the guidance may well be reasonable, it would be helpful to more clearly acknowledge that the proposed thresholds (e.g., survival >0.95 or <0.05 , fewer than 20 individuals per stage) should be interpreted as heuristics derived from a restricted and taxonomically narrow dataset rather than formally validated decision rules. Similarly, the conclusion that integrated metrics such as λ are “often robust” would benefit from additional qualification. Given the structure of the analysis, it is difficult to assess situations in which λ may be more sensitive to sampling issues, and it is possible that such cases are underrepresented in the current dataset. Clarifying these scope limitations would help align the strength of the recommendations with the empirical evidence presented.

Minor Comments

1. Table 1 needs confidence intervals. The variance-component ratios are point summaries derived from bootstrap distributions, but no uncertainty is reported around them. Given that the paper's central message is about the importance of uncertainty, it is somewhat inconsistent to present these key summary statistics without intervals.

2. The shape metric is not self-explanatory. The paper uses the Baudisch and Stott (2019) shape metric, which will not be familiar to all Methods in Ecology and Evolution readers. The current description (a value between -0.5 and $+0.5$, with 0 indicating constant mortality) is minimally sufficient, but a sentence or two on the biological intuition would help readers interpret Figures 3 and 4 without needing to consult the original reference. For example, clarifying what a shape of $+0.3$ versus -0.2 implies for how a plant ages would make the results more immediately interpretable.

3. The quasi-stable distribution truncation rule is consequential but briefly justified. The decision to truncate survivorship trajectories at the point where the cohort reaches within 1% of its quasi-stable stage distribution is noted but not justified. Given that this truncation directly affects the shape estimates, and given that the paper's main findings hinge on shape being sensitive to sampling uncertainty, readers deserve some sensitivity analysis showing how results change if a different threshold (e.g., 5% or 0.5%) is used.